1. Assignment 1: states, rotations and control

A1 Q1.1–Q1.2

A. From a qubit ket to its Bloch vector

Q1.1(a): prove the density-operator/Bloch relation. Write the state, form its outer product, then compare coefficients of the Pauli matrices:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \mathrm{e}^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

$$\boxed{\rho = |\psi\rangle\langle\psi| = \frac{1}{2}(\mathbb{I} + \boldsymbol{r}\cdot\boldsymbol{\sigma})}, \qquad \boldsymbol{r} = \begin{pmatrix}\sin\theta\cos\phi\\ \sin\theta\sin\phi\\ \cos\theta\end{pmatrix}.$$

$$\rho = \frac{1}{2}\begin{pmatrix}1+r_z & r_x - ir_y\\ r_x + ir_y & 1-r_z\end{pmatrix}, \quad r_i = \mathrm{Tr}(\rho\sigma_i).$$

How to prove: use $\cos^2(\theta/2) = (1+\cos\theta)/2$ and $2\cos(\theta/2)\sin(\theta/2) = \sin\theta$. Pure states have $|\boldsymbol{r}| = 1$; mixed qubits have $|\boldsymbol{r}| \le 1$.

B. Pauli algebra and exponentiation

Q1.1(b)–(c): the two standard proof shortcuts.

$$\boxed{\sigma_i\sigma_j = \delta_{ij}\mathbb{I} + i\sum_k \epsilon_{ijk}\sigma_k}, \quad \{\sigma_i,\sigma_j\} = 2\delta_{ij}\mathbb{I}.$$

For a *unit* vector $\boldsymbol{n}$, $A = \boldsymbol{n}\cdot\boldsymbol{\sigma}$ satisfies $A^2 = \mathbb{I}$. Separate even and odd terms of $\mathrm{e}^{i\vartheta A}$:

$$\boxed{\mathrm{e}^{\pm i\vartheta A} = \mathbb{I}\cos\vartheta \pm iA\sin\vartheta}.$$

Approach: check the square first; this works for a two-qubit Pauli product as well (e.g. $Y\otimes Y$).

C. Show a unitary really rotates the Bloch sphere

Q1.1(d) and advanced general-axis rotation.

$$\boxed{R_{\boldsymbol{n}}(\alpha) = \mathrm{e}^{-i\alpha\boldsymbol{n}\cdot\boldsymbol{\sigma}/2}}, \qquad \rho' = R\rho R^\dagger.$$

Use $R\sigma_j R^\dagger$ and read off the new Bloch-vector coefficients. For example,

$$R_x:\ (r_x,r_y,r_z) \mapsto \big(r_x,\ r_y\cos\alpha - r_z\sin\alpha,\\ r_y\sin\alpha + r_z\cos\alpha\big).$$

The analogous cyclic expressions give R_y, R_z. General rotation is described by Rodrigues' formula:

$$\boxed{\boldsymbol{r}' = \boldsymbol{r}\cos\alpha + (\boldsymbol{n}\times\boldsymbol{r})\sin\alpha + \boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{r})(1-\cos\alpha)}.$$

A positive sign in $\exp(+i\alpha\boldsymbol{n}\cdot\boldsymbol{\sigma}/2)$ gives the opposite active rotation under the convention $\rho' = R\rho R^\dagger$. State your sign convention.

D. Hamiltonian → rotation axis → pulse time

Q1.2(a)–(c): recognise the X and Z controls.

$$H = \hbar(t_c X + \Delta Z), \quad \omega = \sqrt{t_c^2 + \Delta^2},$$

$$\boxed{U(\tau) = \cos(\omega\tau)\mathbb{I} - i\sin(\omega\tau)\frac{t_c X + \Delta Z}{\omega}}.$$

The Bloch rotation angle is $\alpha = 2\omega\tau$. For X_α, set $\Delta = 0$; for Z_α, set $t_c = 0$. Hence

$$\boxed{\tau_X = \frac{\alpha}{2t_c}, \qquad \tau_Z = \frac{\alpha}{2\Delta}}.$$

At the specified maxima $t_c/2\pi = 5$ GHz and $\Delta/2\pi = 40$ GHz,

$$\tau_{X_\pi}^{\min} = 50\,\mathrm{ps}, \qquad \tau_{Z_\pi}^{\min} = 6.25\,\mathrm{ps}.$$

Use Euler decompositions for other axes, e.g.

$$R_y(\alpha) = R_z(\pi/2)R_x(\alpha)R_z(-\pi/2),$$

with matrix order as written (the rightmost pulse acts first). An explicit Hadamard synthesis, up to a global phase, is

$$\boxed{H \sim R_z(\pi/2)R_x(\pi/2)R_z(\pi/2)}.$$

E. Timing diagrams with 1 ns resolution

Q1.2(d) and advanced conditional control.

1. Rewrite every target gate as an ordered sequence of X_α and Z_α rotations; rightmost matrix acts first.
2. Choose an *integer* number m of 1 ns slots; set $t_c = \alpha/(2m\,\mathrm{ns})$ or $\Delta = \alpha/(2m\,\mathrm{ns})$ (if within the hardware bounds).
3. Draw horizontal levels for $t_c(t)$, $\Delta(t)$, and the interaction switch $C(t)$; label durations and accumulated angles.
4. At the measurement stage choose $t_c = 0$, so a Z Hamiltonian does not change computational-basis outcome probabilities.

Concrete pulse: an X_π rotation in one 1 ns step needs $t_c/(2\pi) = 0.25$ GHz (not the 5 GHz maximum). The 1 ns clock constrains duration, not the available analogue amplitude.

F. Conditional ZZ phase and CNOT

Q1.2 Part B: two control-state-dependent shifts. With interaction on ($C = 1$) and $t_c = 0$, the target has Z rate $\Delta + \delta_{0,1}$ according to control $|0\rangle, |1\rangle$. Their *difference* produces the entangling phase:

$$\boxed{\chi = 2(\delta_1 - \delta_0)\tau \pmod{2\pi}}.$$

For $\delta_1 - \delta_0 = 2\pi\,50$ MHz, $\chi = \pi$ at $\tau = 5$ ns. Correct the additional *local* Z phases to obtain CZ; then

$$\boxed{\mathrm{CNOT} = (\mathbb{I}\otimes H)\,\mathrm{CZ}\,(\mathbb{I}\otimes H)}.$$

G. No-cloning theorem in three lines

Q1.1 Part B(b). Suppose a unitary clones $|u\rangle$ and $|v\rangle$. Inner products are preserved, but those of two copies square:

$$\langle u|v\rangle = \langle u|v\rangle^2 \Rightarrow \langle u|v\rangle \in \{0,1\}.$$

Nonorthogonal, distinct states violate this: *linearity/unitarity forbids universal cloning.*

2. Assignment 1: entanglement and fidelity

A1 Q1.3–Q1.4

A. Schmidt decomposition: how to find it

Q1.3(a)–(d): pair orthonormal states across a bipartition. Given

$$|\Psi\rangle = \tfrac{1}{\sqrt{2}}(|\alpha_1\rangle_A |0\rangle_B + |\alpha_2\rangle_A |1\rangle_B),$$

$$|\alpha_1\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle\,, \quad |\alpha_2\rangle = \cos\theta\,|0\rangle - \sin\theta\,|1\rangle\,,$$

collect terms by Alice's computational basis and introduce $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$:

$$\boxed{|\Psi\rangle = \cos\theta\,|0\rangle_A |+\rangle_B + \sin\theta\,|1\rangle_A |-\rangle_B}.$$

Absorb any negative coefficient into its associated ket; the *Schmidt coefficients* are $|\cos\theta|, |\sin\theta|$.

General algorithm: (1) write the amplitudes as a matrix M_{ij}; (2) find the singular-value decomposition $M = U\Sigma V^\dagger$; (3) squared singular values λ_j are eigenvalues of ρ_A and ρ_B. Schmidt bases are unique up to phases except within degenerate coefficients; zero-coefficient subspaces are unconstrained.

B. Partial trace and von Neumann entropy

$$\rho_B = \mathrm{Tr}_A\,|\Psi\rangle\langle\Psi| = \cos^2\theta\,|+\rangle\langle+| + \sin^2\theta\,|-\rangle\langle-|\,,$$

$$\boxed{\rho_B = \frac{1}{2}\begin{pmatrix} 1 & \cos 2\theta \\ \cos 2\theta & 1 \end{pmatrix}}.$$

$$\boxed{S(\rho_B) = -\sum_j \lambda_j \log_2 \lambda_j}, \qquad \lambda_\pm = \cos^2\theta, \sin^2\theta.$$

Check: $S = 0$ for product states; $S = 1$ bit for $|\cos\theta| = |\sin\theta|$.

C. Unambiguous state discrimination (USD)

Q1.3 advanced: use positivity and no signalling. The two possible Alice states overlap by

$$s = \langle\alpha_1|\alpha_2\rangle = \cos 2\theta.$$

Let p_1, p_2 be success probabilities conditioned on the equiprobable states. Because Alice's local action cannot alter Bob's unconditional state, the unnormalised Bob "failure" operator must be positive:

$$\rho_B^{\text{fail}} = \tfrac{1}{2}\begin{pmatrix} 1-p_1 & s \\ s & 1-p_2 \end{pmatrix} \succeq 0.$$

Thus $(1-p_1)(1-p_2) \geq |s|^2$. By AM–GM,

$$\boxed{P_{\text{success}} = \tfrac{p_1+p_2}{2} \leq 1 - |s| = 1 - |\cos 2\theta|}.$$

This is tight for equal priors. USD may have an inconclusive result but *never* falsely identifies a state.

D. Teleportation through a nonmaximal Bell pair

Q1.4 Part A: calculate the output with outcome weights. Input $|a\rangle = \alpha\,|0\rangle + \zeta\,|1\rangle$ with $\alpha = \cos(\theta/2)$, $\zeta = \mathrm{e}^{i\phi}\sin(\theta/2)$. Resource:

$$|AB\rangle = \cos\beta\,|01\rangle - \sin\beta\,|10\rangle\,.$$

Do this in order:

1. Expand $|a\rangle \otimes |AB\rangle$ in the (a, A, B) ordering.
2. Project Alice's two systems onto $\Phi^\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $\Psi^\pm = (|01\rangle \pm |10\rangle)/\sqrt{2}$.
3. Apply Bob's outcome-specific Pauli correction. The *unnormalised* corrected state for either Φ outcome is $\frac{1}{\sqrt{2}}(\alpha\cos\beta\,|0\rangle + \zeta\sin\beta\,|1\rangle)$; for either Ψ outcome it is $\frac{1}{\sqrt{2}}(\alpha\sin\beta\,|0\rangle + \zeta\cos\beta\,|1\rangle)$, up to a global phase.
4. Sum corrected density operators, retaining outcome probabilities automatically.

$$\boxed{\rho_{\text{out}} = \begin{pmatrix} |\alpha|^2 & \alpha\zeta^* \sin 2\beta \\ \alpha^*\zeta \sin 2\beta & |\zeta|^2 \end{pmatrix}}.$$

This is a dephasing channel on the input Bloch sphere: z population is preserved; transverse coherence is reduced by $\sin 2\beta$.

E. Fidelity, then average over unknown inputs

For a pure target and mixed output,

$$F = \langle a|\,\rho_{\text{out}}\,|a\rangle\,.$$

Substitute α, ζ:

$$\boxed{F(\theta,\phi) = 1 - \tfrac{1}{2}(1 - \sin 2\beta)\sin^2\theta}.$$

There is no ϕ dependence. Uniform Bloch-sphere averaging uses the *area element* $\sin\theta\,d\theta\,d\phi$:

$$\bar{F} = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} F\sin\theta\,d\theta\,d\phi,$$

$$\boxed{\bar{F} = \frac{2 + \sin 2\beta}{3}}.$$

Checks: $\beta = \pi/4 \Rightarrow F = 1$; $\beta = 0 \Rightarrow \bar{F} = 2/3$, the classical limit for an unknown qubit.

F. Werner-state teleportation

Q1.4 Part B: exploit linearity instead of redoing Bell projections.

$$\rho_W = (1-p)\,|\Psi^-\rangle\langle\Psi^-| + \tfrac{p}{4}\mathbb{I}_4.$$

The Bell resource teleports perfectly; the fully mixed part produces $\mathbb{I}_2/2$. Therefore

$$\boxed{\rho_{\text{out}} = (1-p)\,|a\rangle\langle a| + \frac{p}{2}\mathbb{I}_2, \quad F = 1 - \frac{p}{2}}.$$

Exam habit: distinguish a *conditional normalised* output (one measurement result) from the *average* teleported state (sum over all four results). The latter is used for the usual fidelity.

3. Assignment 2: open systems and QEC

A2 Q1–Q2

A. Lindblad equation → Bloch differential equations

Q1 Part A: arbitrary initial state, steady state, plots, purity.

$$\dot\rho = -\frac{i\Omega}{2}[X,\rho] + \gamma\mathcal{D}[Z]\rho, \quad \mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}.$$

Insert $\rho = (\mathbb{I} + xX + yY + zZ)/2$. Compare the X, Y, Z coefficients (or use $\dot r_j = \mathrm{Tr}(\sigma_j \dot\rho)$):

$$\boxed{\dot x = -2\gamma x, \quad \dot y = -2\gamma y - \Omega z, \quad \dot z = \Omega y}.$$

For $\Omega > \gamma > 0$, let $\omega = \sqrt{\Omega^2 - \gamma^2}$. With (x_0, y_0, z_0) arbitrary,

$$\begin{aligned} x(t) &= x_0 \mathrm{e}^{-2\gamma t}, \\ y(t) &= \mathrm{e}^{-\gamma t}\left[y_0 \cos\omega t - \frac{\gamma y_0 + \Omega z_0}{\omega}\sin\omega t\right], \\ z(t) &= \mathrm{e}^{-\gamma t}\left[z_0 \cos\omega t + \frac{\gamma z_0 + \Omega y_0}{\omega}\sin\omega t\right]. \end{aligned}$$

Workflow: derive ODEs → solve $\dot{\boldsymbol{r}} = A\boldsymbol{r}$ → set $\dot{\boldsymbol{r}} = 0$ → recover $\rho(t)$ → plot $\boldsymbol{r}(t)$ and purity:

$$\rho_{\mathrm{ss}} = \mathbb{I}/2 \quad (\Omega, \gamma > 0), \qquad \boxed{\mathrm{Tr}\,\rho^2 = \frac{1 + x^2 + y^2 + z^2}{2}}.$$

For $|0\rangle$ initially, $(x_0, y_0, z_0) = (0, 0, 1)$. If $\Omega \le \gamma$, use real exponential/hyperbolic solutions instead of oscillatory sines. Pure dephasing with $\Omega = 0$ leaves z conserved.

B. Damped atom → Kraus representation

Q1 Part B: generalized amplitude damping. For $|g\rangle = (1, 0)^T$, $|e\rangle = (0, 1)^T$ define

$$\Gamma_\downarrow = \tfrac{\gamma}{2}(n+1), \quad \Gamma_\uparrow = \tfrac{\gamma}{2}n, \quad \Gamma_1 = \tfrac{\gamma}{2}(2n+1).$$

Then $x, y \propto \mathrm{e}^{-\Gamma_1 t/2}$ and

$$\boxed{z(t) = z_\infty + (z_0 - z_\infty)\mathrm{e}^{-\Gamma_1 t}, \quad z_\infty = \frac{1}{2n+1}}.$$

Set $q = (n+1)/(2n+1)$ (ground equilibrium probability), $\lambda = 1 - \mathrm{e}^{-\Gamma_1 t}$:

$$\begin{aligned} K_0 &= \sqrt{q}\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, & K_1 &= \sqrt{q}\begin{pmatrix} 0 & \sqrt{\lambda} \\ 0 & 0 \end{pmatrix}, \\ K_2 &= \sqrt{1-q}\begin{pmatrix} \sqrt{1-\lambda} & 0 \\ 0 & 1 \end{pmatrix}, & K_3 &= \sqrt{1-q}\begin{pmatrix} 0 & 0 \\ \sqrt{\lambda} & 0 \end{pmatrix}. \end{aligned}$$

Check $\sum_j K_j^\dagger K_j = \mathbb{I}$, $t = 0$ gives identity and $t \to \infty$ gives thermal equilibrium.

C. The Steane code: start with the parity checks

Q2 Part A: Tanner graph, logicals, distance, syndromes.

$$\boxed{[[n, k, d]] = [[7, 1, 3]], \qquad k = n - r = 7 - 6 = 1.}$$

The three X-check supports are $\{4,5,6,7\}$, $\{2,3,6,7\}$, $\{1,3,5,7\}$; the Z checks have the same supports. The classical parity-check matrix is

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Tanner graph: draw one vertex for each physical qubit, another for each parity check, and an edge wherever $H_{ij} = 1$.

All same-Pauli checks commute. Opposite-type checks overlap in 4 (same support) or 2 (different supports) locations, hence an *even* number of XZ anticommutations, so they commute.

D. General syndrome-lookup algorithm

The syndrome of X_j is the jth column of H, measured by the Z *checks*; swap roles for Z_j:

syndrome	000	001	010	011	100	101	110	111
qubit	$--$	1	2	3	4	5	6	7

Steps: measure three Z checks → correct X_j; measure three X checks → correct Z_k. A $Y_j = iX_jZ_j$ generates *both* signatures. The syndrome measures an error without observing logical amplitudes.

Logical representatives $X_L = X^{\otimes 7}$, $Z_L = Z^{\otimes 7}$ commute with all checks and anticommute with each other. Multiply by stabilisers:

$$X_L(\mathbb{III}XXXX) = X_1X_2X_3,$$

$$Z_L(\mathbb{III}ZZZZ) = Z_1Z_2Z_3.$$

Thus $d \le 3$, while distinct nonzero single-error syndromes rule out nontrivial weight-1 or weight-2 logicals; hence $d = 3$.

E. Failure probability and transversal CNOT

Q2 Part A(4)–(5), Part B(a)–(b). For a bit-only independent-error rate p, an *upper bound* follows from requiring at most one error:

$$\boxed{P_{\mathrm{fail}} \le 1 - (1-p)^7 - 7p(1-p)^6}.$$

The exact lookup decoder failure probability is

$$\boxed{\begin{aligned} P_{\mathrm{fail}} = {} & 21p^2(1-p)^5 + 7p^3(1-p)^4 \\ & + 28p^4(1-p)^3 + 7p^6(1-p) + p^7. \end{aligned}}$$

For independent X, Z sectors, sum sector-wise upper bounds (or use independence to bound by $1 - P_{\le 1}(p_X)P_{\le 1}(p_Z)$). For a transversal logical CNOT, apply seven physical $\mathrm{CNOT}_{c_j \to t_j}$ gates and check Heisenberg transformations:

$$\boxed{X_L^c \mapsto X_L^c X_L^t, \qquad Z_L^t \mapsto Z_L^c Z_L^t},$$

with Z_L^c, X_L^t unchanged.

4. Assignment 3: blockade and ion pulses

A3 Q1–Q2

A. Rydberg blockade → controlled X rotation

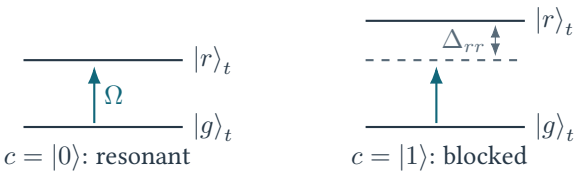
Q1.1–Q1.5: control on $|0\rangle$, not $|1\rangle$. Here $|0\rangle \equiv |g\rangle$, $|1\rangle \equiv |r\rangle$, and the target is driven resonantly with angular Rabi frequency Ω. If control is $|0\rangle$, the target rotates; if control is already $|r\rangle$, double excitation is off resonance:

$$U_x(\theta) = \mathrm{e}^{-i\theta X/2} = \begin{pmatrix} c & -is \\ -is & c \end{pmatrix}, \quad c = \cos\frac{\theta}{2},\ s = \sin\frac{\theta}{2}.$$

$$\boxed{CU_x(\theta) = \begin{pmatrix} c & -is & 0 & 0 \\ -is & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},}$$

with ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (control first). Here $\theta = \Omega\tau$.

Physical sequence: initialise neighbouring atoms; shine the resonant laser at the target for τ; in the $|0\rangle_c$ sector it performs Rabi oscillations, whereas in the $|1\rangle_c$ sector the interaction shifts $|rr\rangle$ and suppresses the target excitation.



The high-fidelity requirement is $|\Delta_{rr}| \gg \Omega$ if both are angular frequencies (or $|U_{rr}| \gg \hbar\Omega$ in energy units). Scalability is limited by finite-range blockade, crosstalk and Rydberg decay.

B. Turn CU_x into an exact controlled NOT

Q1.6–Q1.7: conditional phase matters. Choosing $\theta = 3\pi$ yields $U_x(3\pi) = iX$, hence the control-0 sector picks up a relative phase i. Prepend $R_z(\pi/2)$ on the *control*:

$$\boxed{CU_x(3\pi)\,[R_z(\pi/2) \otimes \mathbb{I}] = \mathrm{e}^{i\pi/4}\, C_0X},$$

where C_0X flips the target only if control is $|0\rangle$. Conjugate with an X on the control to obtain standard control-on-1 CNOT.

$$\boxed{\tau = \frac{3\pi}{\Omega}; \qquad \frac{\Omega}{2\pi} = 2\,\mathrm{kHz} \Rightarrow \tau = 0.75\,\mathrm{ms}.}$$

A π target pulse gives $-iX$, *not* X. Its phase is conditional and requires local correction. Always distinguish pulse *area* θ from duration τ.

C. Trapped-ion sidebands: track motion explicitly

Q2 Part A: $\pi_A, 2\pi_B, \pi_A$ gate. Use $|A\,B, n\rangle$ with internal g/e and shared phonon number n. A red sideband couples

$$\boxed{|e, n\rangle \leftrightarrow |g, n+1\rangle},$$

with a π pulse contributing $-i$ each way, while $|g, 0\rangle$ is dark. On ion B the auxiliary transition is $|g, 1\rangle \leftrightarrow |a, 0\rangle$: a 2π pulse returns it with sign -1.

Input	after π_A	after $2\pi_B$	after π_A
$\lvert gg, 0\rangle$	$\lvert gg, 0\rangle$	$\lvert gg, 0\rangle$	$\lvert gg, 0\rangle$
$\lvert ge, 0\rangle$	$\lvert ge, 0\rangle$	$\lvert ge, 0\rangle$	$\lvert ge, 0\rangle$
$\lvert eg, 0\rangle$	$-i\,\lvert gg, 1\rangle$	$+i\,\lvert gg, 1\rangle$	$+\lvert eg, 0\rangle$
$\lvert ee, 0\rangle$	$-i\,\lvert ge, 1\rangle$	$-i\,\lvert ge, 1\rangle$	$-\lvert ee, 0\rangle$

$$\boxed{U_{AB} = \mathrm{diag}(1, 1, 1, -1) = CZ}.$$

Method: write the effect of *each* pulse on each basis input, include the phonon and phase, and only then build the 4×4 matrix. The shared mode must end in $|0\rangle$.

D. From CZ to CNOT; from CZ to CCZ

$$\boxed{\mathrm{CNOT}_{A\to B} = (\mathbb{I} \otimes H)CZ(\mathbb{I} \otimes H)}.$$

For three ions A, B, C, add a red-sideband π pulse to map the excitation of a second control (e.g. ion C) into the motional bus, then the auxiliary 2π conditional phase, then undo the mappings in reverse. The five-pulse pattern is schematically

$$\pi_A\ \pi_C\ 2\pi_B\ \pi_C\ \pi_A.$$

With appropriate sideband addressing and phase conventions, only $|eee\rangle$ acquires a negative sign:

$$\boxed{CCZ = \mathrm{diag}(1, 1, 1, 1, 1, 1, 1, -1)}.$$

Conjugate by H on target B to obtain the Toffoli (CCNOT). The exact pulse design must ensure the bus distinguishes the intended control sector and returns to $n = 0$.

E. Generic hardware-gate answer template

1. Declare computational basis and any auxiliary states/modes.
2. Specify the resonant/detuned transition and why it is driven or dark.
3. Translate pulse area $\Theta = \Omega_{\mathrm{eff}}\tau$ into $-i$ for π and -1 for 2π within the coupled two-state subspace.
4. Track relative phases; confirm all population returns to the computational manifold.
5. Build the gate and state necessary single-qubit corrections.

5. Assignment 3: circuits, photons and lookup

A3 Q3–Q4 + complete index

A. Cooper-pair box: start with $\mathcal{L} = T - V$

Q3.1–Q3.4: derive instead of guessing the Hamiltonian. Node flux Φ is the generalised coordinate, with voltage $\dot{\Phi}$. The Josephson potential is $-E_J \cos\varphi$, where $\varphi = 2\pi\Phi/\Phi_0$ and $\Phi_0 = h/(2e)$. With $C_\Sigma = C_0 + C_c$,

$$\boxed{\mathcal{L} = \tfrac{1}{2}C_0\dot{\Phi}^2 + \tfrac{1}{2}C_c(\dot{\Phi} - V_g)^2 + E_J\cos\varphi}.$$

Standard four-step method:

1. Compute conjugate charge: $Q = \partial\mathcal{L}/\partial\dot{\Phi} = C_\Sigma\dot{\Phi} - C_cV_g$.
2. Solve $\dot{\Phi} = (Q + C_cV_g)/C_\Sigma$.
3. Legendre transform $H = Q\dot{\Phi} - \mathcal{L}$, dropping only terms constant in Q, Φ:

$$\boxed{H = \frac{(Q + C_cV_g)^2}{2C_\Sigma} - E_J\cos\varphi}.$$

4. Quantise $[\Phi, Q] = i\hbar$, then write $H\psi = E\psi$ (in flux representation, $Q = -i\hbar\partial_\Phi$).

For $V_g = 0$ near a Josephson minimum, $\cos\varphi \simeq 1 - \varphi^2/2$ and

$$\boxed{\hbar\omega_p = \sqrt{8E_JE_C}, \qquad E_C = \frac{e^2}{2C_\Sigma}},$$

$$\boxed{E_m \simeq -E_J + \hbar\omega_p\left(m + \tfrac{1}{2}\right)}.$$

Therefore $E_1 - E_0 \simeq \sqrt{8E_JE_C}$. The nonquadratic cosine produces weak anharmonicity (the resource for selectively addressing the $|0\rangle \to |1\rangle$ transition).

B. Numerical spectrum: charge-basis matrix

Q3.5–Q3.6: truncate and diagonalise. Define $n = Q/(2e)$ and $n_g = -C_cV_g/(2e)$. Then

$$\boxed{H = 4E_C(n - n_g)^2 - E_J\cos\varphi}.$$

In the integer Cooper-pair basis, $e^{\pm i\varphi}|n\rangle = |n \pm 1\rangle$, giving the tridiagonal matrix

$$\boxed{H_{nm} = 4E_C(n - n_g)^2\delta_{nm} - \frac{E_J}{2}(\delta_{n,m+1} + \delta_{n,m-1})}.$$

Approach: choose symmetric cutoff $n = -N, \ldots, N$; compute lowest four eigenvalues vs n_g; increase N until results converge; minimise $E_1(n_g) - E_0(n_g)$ over one period ($n_g \in [0, 1]$). The smallest gap is near half-integer offset charge (the charge-degeneracy point). For an exact numerical value use the specified capacitance and E_J ratios, not the harmonic approximation.

C. Beam splitters, Bell states, HOM and MZI

Q4 Part A/B: transform creation operators *rather than guessing photon paths.* Choose one balanced-beam-splitter convention and use it consistently for both polarisations:

$$a_s^\dagger \mapsto \frac{c_s^\dagger + d_s^\dagger}{\sqrt{2}}, \quad b_s^\dagger \mapsto \frac{c_s^\dagger - d_s^\dagger}{\sqrt{2}}, \qquad s = H, V.$$

Calculation algorithm: (1) rewrite a Bell ket using $a_H^\dagger a_V^\dagger|\text{vac}\rangle$ etc.; (2) replace the operators with their outputs; (3) commute bosonic operators; (4) convert $(c^\dagger)^2|\text{vac}\rangle = \sqrt{2}|2\rangle$; (5) group and normalise output states.

For the *antisymmetric* singlet input,

$$\boxed{|\Psi^-\rangle_{ab} = \frac{a_H^\dagger b_V^\dagger - a_V^\dagger b_H^\dagger}{\sqrt{2}}|\text{vac}\rangle},$$

all bunched terms cancel, giving a coincidence of one photon per output port (up to an overall phase):

$$|\Psi^-\rangle_{ab} \to -|\Psi^-\rangle_{cd}.$$

For symmetric inputs $|\Psi^+\rangle, |\Phi^\pm\rangle$, coincidence terms cancel and photons bunch; the H/V content of the bunch differs between Bell states. For instance, with our convention,

$$\boxed{|\Psi^+\rangle \to \frac{|HV, 0\rangle - |0, HV\rangle}{\sqrt{2}}}.$$

The relative sign depends on the beam-splitter convention and which input/output ports are labelled a, b, c, d.

In a balanced Mach–Zehnder interferometer, both arms undergo another common-mode optical unitary. The singlet's antisymmetric structure is invariant up to phase, so it still gives one count at each output detector. This is *HOM-type two-photon interference* (antibunching of the antisymmetric state), not ordinary HOM bunching. A coincidence alone does not prove indistinguishability: vary arrival-time delay experimentally.

D. Five-page map of *every assignment*

Question	Method to locate	Page
A1 Q1.1	Bloch, Pauli, rotations; no cloning	1
A1 Q1.2	Pulse amplitude, timing, controlled phase	1
A1 Q1.3	Schmidt, trace, entropy, USD bound	2
A1 Q1.4	Bell projection, fidelity, Werner state	2
A2 Q1	Bloch ODEs, steady state, thermal Kraus	3
A2 Q2	Steane Tanner graph, lookup, failure, CNOT	3
A3 Q1	Blockade, controlled U_x, pulse phase	4
A3 Q2	Red-sideband pulses, CZ and CCZ	4
A3 Q3	CPB Lagrangian, spectrum, gap	5
A3 Q4	Bell-state optics, MZI, HOM	5

Last check: specify basis and ordering; distinguish angular Ω from $\Omega/(2\pi)$; keep conditional phases; include phonon numbers; normalise Fock states; test $\text{Tr}\,\rho = 1$, $\sum K_j^\dagger K_j = \mathbb{I}$ and $U^\dagger U = \mathbb{I}$. For full derivations use the three submitted assignment solution PDFs.